Homework 3

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Part 1. Set Up

```
# Homework Set Up and install packages
library(tinytex)
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.0     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.2     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.1
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
```

```
library(here)
```

```
here() starts at /home/jovyan/ENVS-193DS-Homework-03
```

```
library(janitor)
```

```
Attaching package: 'janitor'

The following objects are masked from 'package:stats':

    chisq.test, fisher.test
```

```
library(readxl)

# creating object and reading in personal data set
```

```r
my_data <- read_csv("personal-data (2).csv",
# editing format of column names
show_col_types = FALSE,
                    col_names = TRUE)

# clean names
my_data_clean <- my_data %>%
  clean_names()
# insert kelp dataset
kelp <- read_csv(here("data", "temp-kelp.csv"))
```

```
Rows: 32 Columns: 2
-- Column specification --------------------------------------------------------
Delimiter: ","
dbl (2): temp_c, kelp_elong

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Part 2. Problems

Problem 1. Giant Kelp Fronds

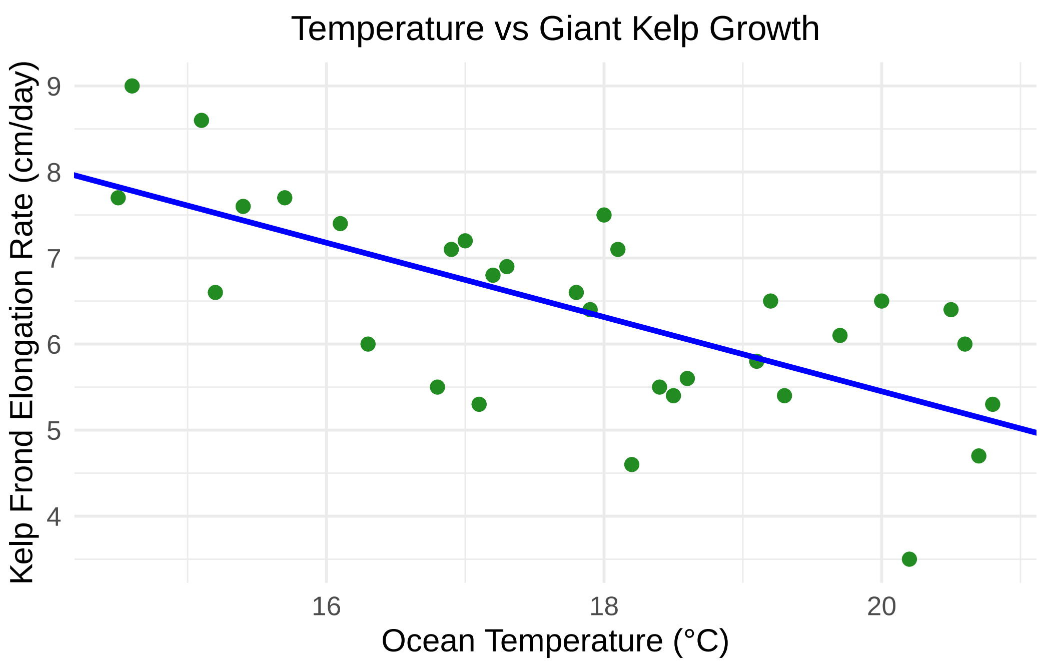
a. Appropriate test

The appropriate tests would include Pearson correlation and linear regression. Pearson correlation measures the strength and direction of the relationship between temperature and kelp elongation rate, while linear regression models how kelp growth changes with temperature

b. Visualization

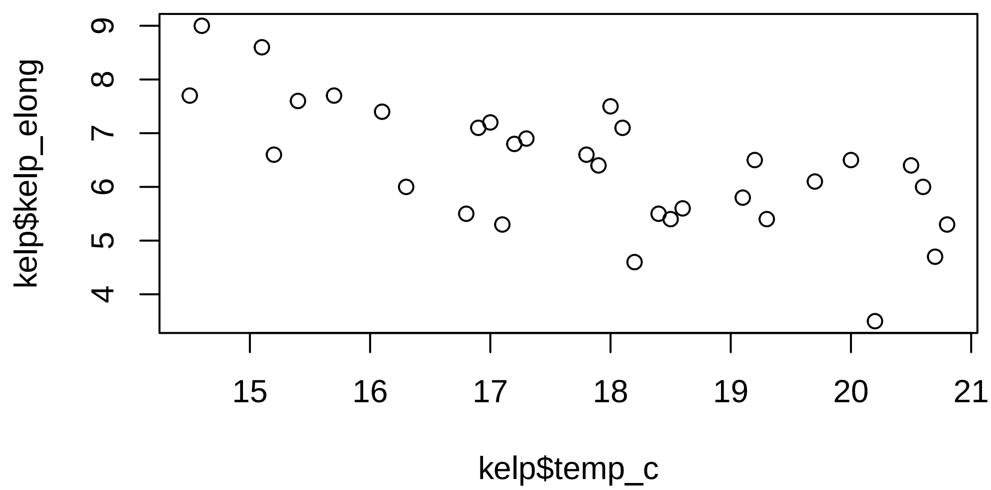
```r
# Fit linear model
kelp_model <- lm(kelp_elong ~ temp_c, data = kelp)

# Create scatterplot + regression line directly from model coefficients
ggplot(kelp, aes(x = temp_c, y = kelp_elong)) +

  # Data points
```

```r
geom_point(color = "forestgreen", size = 2) +

# Regression line (no slope/intercept objects needed)
geom_abline(
  intercept = coef(kelp_model)[1],
  slope = coef(kelp_model)[2],
  color = "blue",
  linewidth = 1
) +

# Labels with units
labs(
  x = "Ocean Temperature (°C)",
  y = "Kelp Frond Elongation Rate (cm/day)",
  title = "Temperature vs Giant Kelp Growth"
) +

# Non-default theme
theme_minimal() +

# Formatting
theme(
  plot.title = element_text(hjust = 0.5),
  axis.title = element_text(size = 12),
  axis.text = element_text(size = 10)
)
```

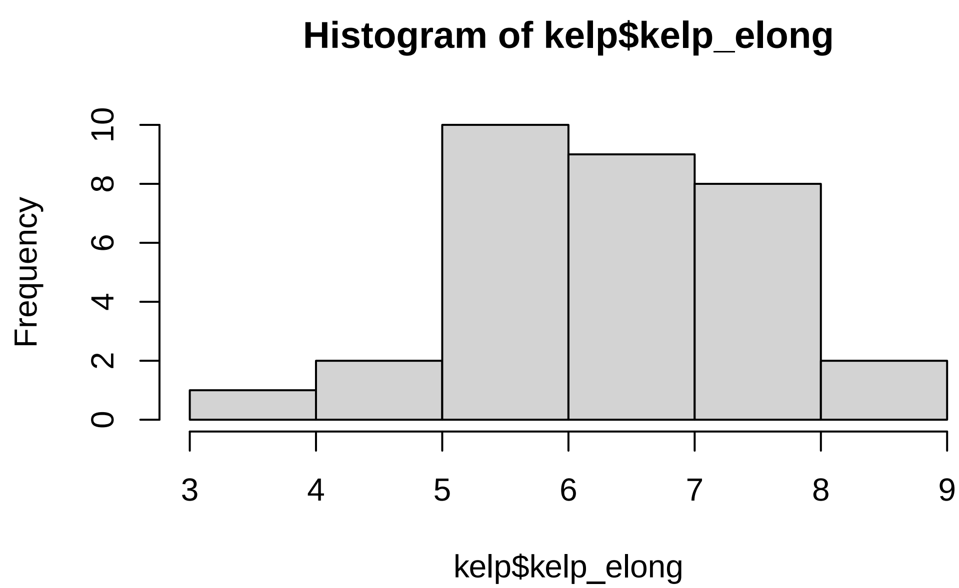


c. Check your assumptions and run test

```
# check relationship
plot(kelp$temp_c, kelp$kelp_elong)
```




```r
# check distribution
hist(kelp$kelp_elong)
```



I checked linearity using a scatterplot and assessed the distribution using a histogram. The relationship appears approximately linear with no significant outliers, and the data appear reasonably normally distributed.

```r
# Run regression
model <- lm(kelp_elong ~ temp_c, data = kelp)
summary(model)
```

```

Call:
lm(formula = kelp_elong ~ temp_c, data = kelp)

Residuals:
    Min      1Q  Median      3Q     Max
-1.8643 -0.5017  0.1790  0.5659  1.2177

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 14.08645    1.49219   9.440 1.72e-10 ***
temp_c      -0.43179    0.08321  -5.189 1.37e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8665 on 30 degrees of freedom
Multiple R-squared:  0.473, Adjusted R-squared:  0.4554
F-statistic: 26.93 on 1 and 30 DF,  p-value: 1.366e-05
```

d. Results communication

A linear regression was used to test the relationship between temperature and kelp growth. The results show a significant negative relationship (p = 1.366e-05, R^2 = 0.473), indicating that as temperature increases, kelp elongation rate decreases. Indicating that temperature does significantly predict kelp elongation rate.

e. Test implications

These results suggest that increasing ocean temperatures may impact kelp growth rates. Since the relationship appears to be negative, warming oceans could reduce kelp productivity, affecting ecosystem health and function. This information can help guide conservation and climate adaptation and management strategies.

f. Double check work

```r
# Pearson correlation
cor.test(kelp$temp_c,kelp$kelp_elong)
```

```
	Pearson's product-moment correlation

data:  kelp$temp_c and kelp$kelp_elong
t = -5.189, df = 30, p-value = 1.366e-05
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.8359665 -0.4460157
sample estimates:
       cor
-0.6877489
```

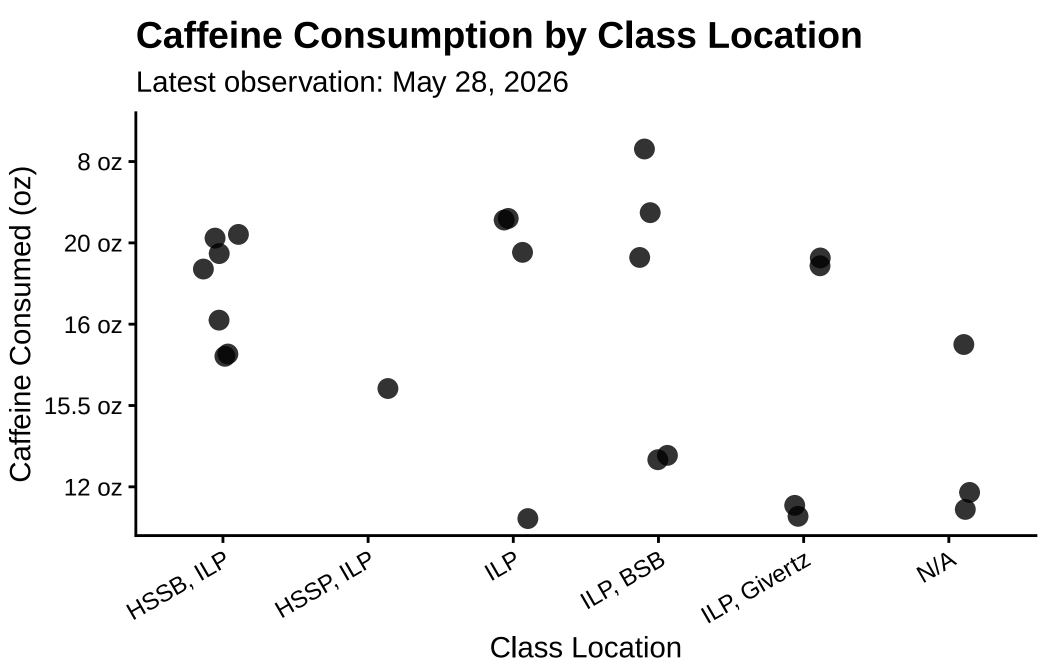
Both Pearson correlation and linear regression led to the same reasonable conclusion about the relationship between temperature and kelp elongation rate. While correlation measures strength and direction, regression additionally quantifies the predictive relationship between variables.

Problem 2. Personal Data

a. Update plots

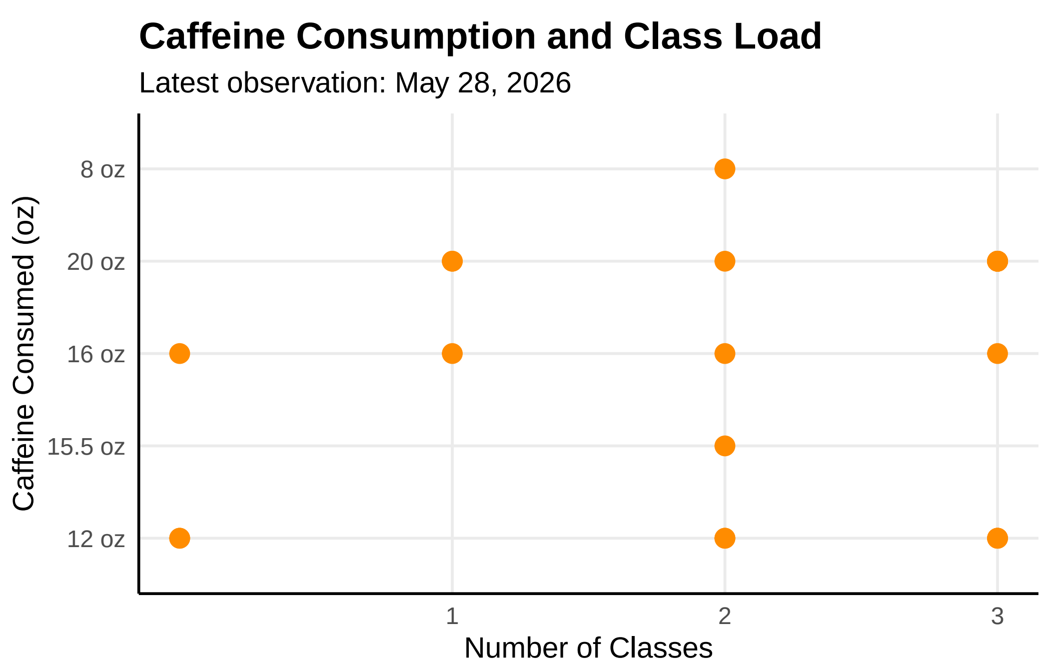
```r
# Updated visualization plots from Homework 2

# updating Plot 1 (Categorical)
ggplot(my_data_clean, aes(x = class_location, y = ounces_consumed, fill = class_location)) +

  # jitter points
  geom_jitter(width = 0.15, size = 3, alpha = 0.8) +

  # custom colors
  scale_fill_manual(values = c("orchid","skyblue","gold","green","purple","tomato")) +

  # labels
```

```r
  labs(
    title = "Caffeine Consumption by Class Location",
    subtitle = "Latest observation: May 28, 2026",
    x = "Class Location",
    y = "Caffeine Consumed (oz)"
  ) +

  # clean theme
  theme_classic() +

  # formatting
  theme(
    legend.position = "none",
    plot.title = element_text(size = 14, face = "bold"),
    axis.text.x = element_text(angle = 30, hjust = 1)
  )
```




```r
# updating Plot 2 (Continuous)

ggplot(my_data_clean, aes(x = number_of_classes, y = ounces_consumed)) +

  # scatter points
```

```r
geom_point(size = 3, color = "darkorange") +

# x-axis formatting
scale_x_continuous(breaks = c(1, 2, 3)) +

# labels
labs(
  title = "Caffeine Consumption and Class Load",
  subtitle = "Latest observation: May 28, 2026",
  x = "Number of Classes",
  y = "Caffeine Consumed (oz)"
) +

# clean theme
theme_minimal() +

# reduce clutter
theme(
  panel.grid.minor = element_blank(),
  axis.line = element_line(color = "black"),
  plot.title = element_text(size = 14, face = "bold")
)
```



b. Captions

Figure 1. Caffeine consumption (oz) varies across class locations. Each point represents an observation day, with jitter applied to reduce overlap. Consumption appears to vary by location, suggesting that where classes are held may influence caffeine intake patterns.

Figure 2. Caffeine consumption (oz) in relation to number of classes attended per day. Each point represents a single day. While there is some variability, higher class loads may be associated with increased caffeine consumption.

Problem 3. Affective visualization

a. Describe in words what an affective visualization could look like for your personal data

An affective visualization of my data would highlight how caffeine consumption changes based on academic workload and environment. Instead of focusing only on numerical values, the visualization would emphasize how different days feel, using color intensity or size to represent higher or lower caffeine intake. For example, days with more assignments due could be represented with darker colors to convey increased stress or fatigue, and the size of the point represented by a coffee cup could be larger to represent days with a higher number of classes. This approach would help communicate not just the data, but the experience behind the collection process.

b. Create a sketch (on paper) of your idea

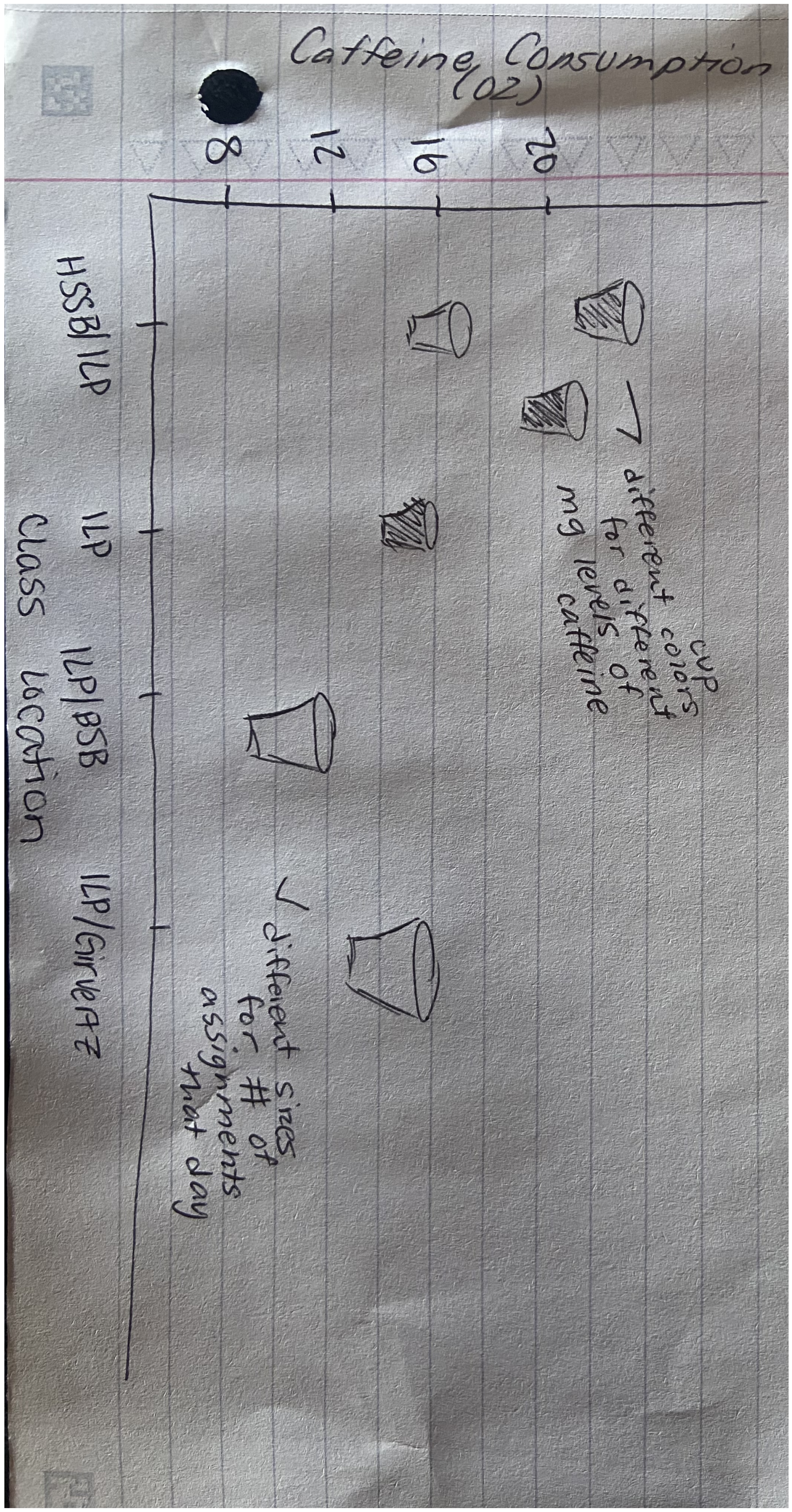


Figure 1: Sketch of personal data visualizaton

c. Make a draft of the visualization

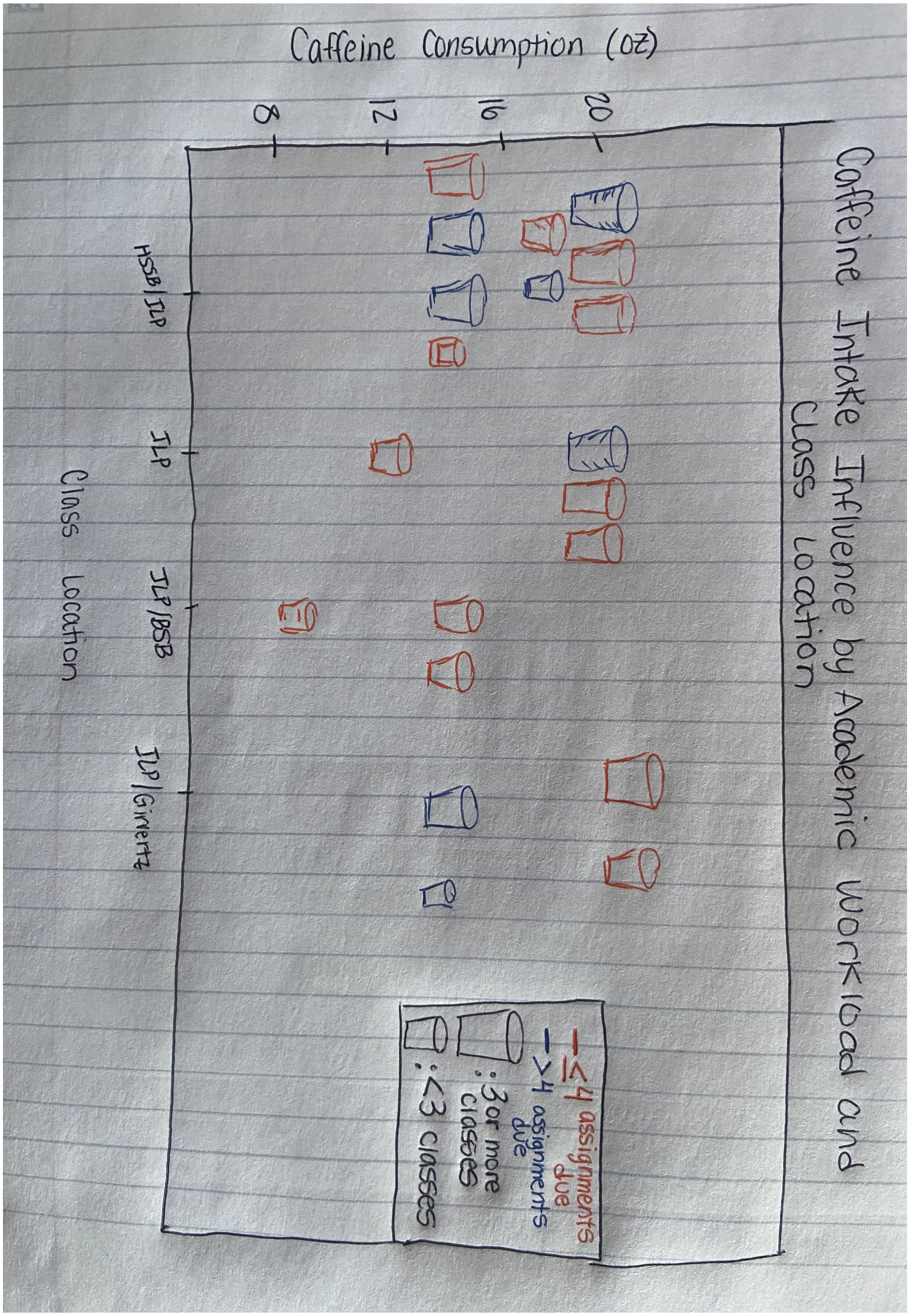


Figure 2: Sketch of personal data visualizaton

d. Write an artist statement

Content

This visualization represents daily caffeine consumption in relation to academic workload and class environment, highlighting how different days influence energy needs. ### Influences I was inspired by projects such as the Dear Data reference project, which incorporated personal story telling into data and statistics, creating a unique visual expression. ### Form The piece is a hand-drawn visualization that uses color and sizing to represent variation in caffeine intake based on class environment and academic workload. ### Process I began by thinking of how I wanted to display the data I have collected, then I decided to begin sketching how to represent my data visually, I choose to use shapes and colors to translate my dataset into a more expressive format using drawing and blending a traditional figure with creative features.

e. Prep materials to share with class

Presentation slides

View my slides

Problem 4. Statistical critique

a. Revisit and summarize

The authors used statistical tests such as linear regression to evaluate relationships between variables and assess how predictors influence response variables.

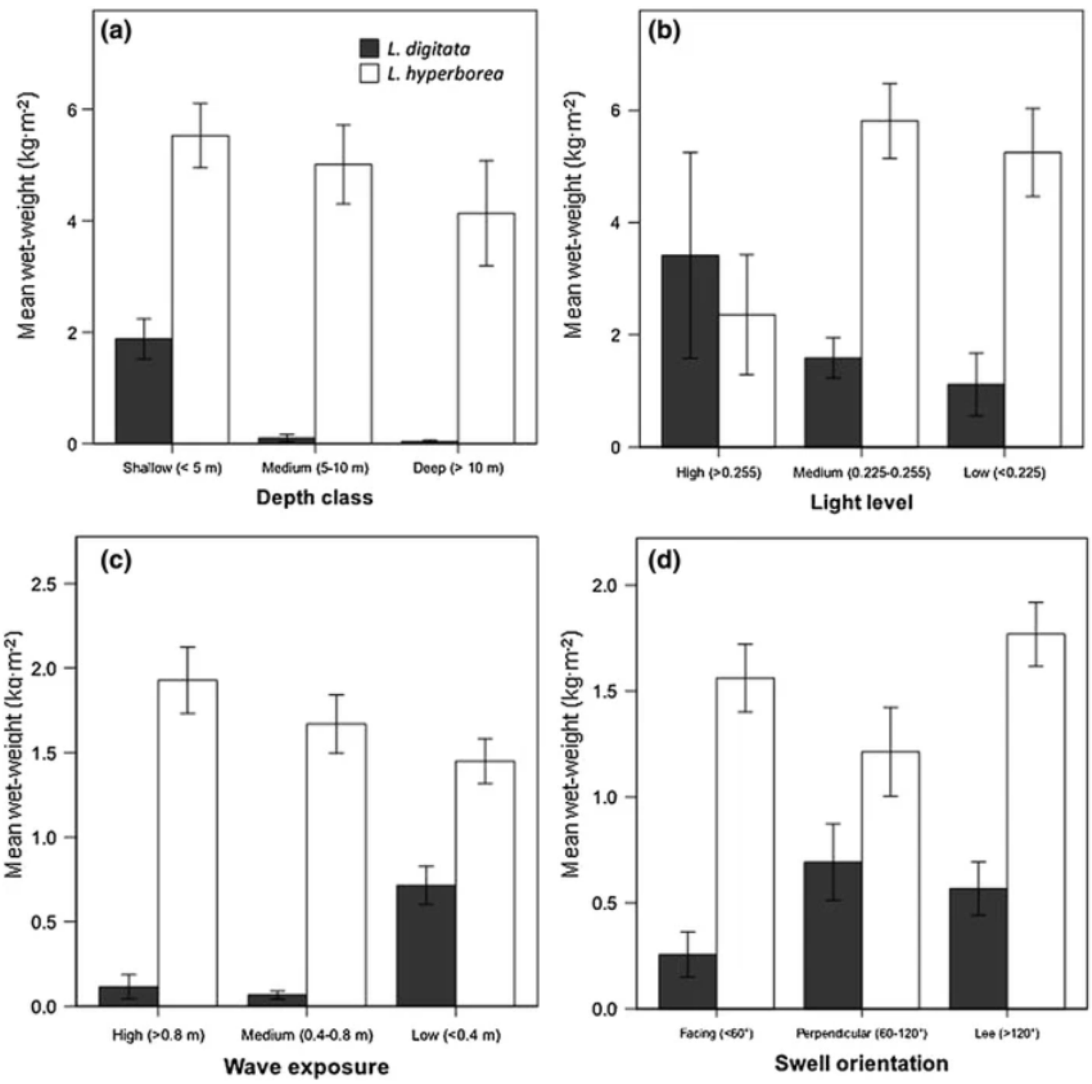


Figure 3: Figure showing kelp biomass differences among environmental groups

b. Visual clarity

The figure presents the data clearly with labeled axes and an approapriate layout. However, while summary statistics are shown, the underlying data points are are not individualized which may limit interpretation.

c. Aesthetic clarity

The figure includes more than just one response and predictor variable and can become confusing. The data to ink ratio could be improved by emphasizing one aspect of the experiment and minimizing background elements.

d. Recommendations

The figure could be improved by reducing visual clutter, such as removing excess gridlines and simplifying the design. Adding raw data points alongside summary statistics would improve interpretability, and more intentional color choices could help distinguish groups more effectively.

“‘